Chapter6: Electric Potential

Textbook: Fundamentals of Physics by Halliday, Resnick & Walker [extended 9th edition]

Topics to be covered:

24-2: Electric Potential Energy, 24-3: Electric Potential, 24-4: Equipotential Surface, 24-5: Calculating the Potential from the Field, 24-6: Potential Due to a Point Charge, 24-7: Potential Due to a Group of Point Charge, 24-8: Potential Due to an Electric Dipole, 24-9: Potential Due to a Continuous Charge Distribution, 24-10: Calculating the Field from the Potential.

Theory

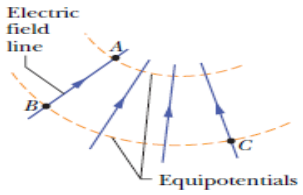
1. Define Electric Potential and equipotential surface.
2. Derive the expression of work done by an applied force to move a particle of charge q from one point to another in an electric field.
3. Derive the general equation for calculating the electric potential from electric field.
4. Calculate electric field from electric potential.
5. Find out the expression for the electric potential due to a point charge.
6. Derive the expression for potential due to an electric dipole at any arbitrary point. Reduce this equation for the point being on the dipole axis.

Problems

Chapter-24

1. [1] A particular 12 V car battery can send a total charge of 3×10^5 C through a circuit, from one terminal to the other. If this entire charge undergoes a change in electric potential of 12 V, how much energy is involved?
2. [2] The electric potential difference between the ground and a cloud in a particular thunderstorm is 1.2×10^9 V. In the unit electron-volts, what is the magnitude of the change in the electric potential energy of an electron that moves between the ground and the cloud?

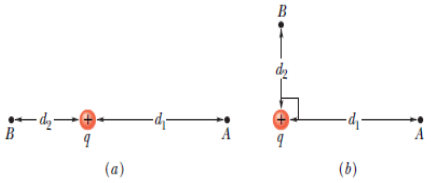
3. [6]When an electron moves from A to B along an electric field line in the figure, the electric field does 3.94×10^{-19} J of work on it. What are the electric potential differences (a) V_B-V_A and (b)$V_C- V_B$?



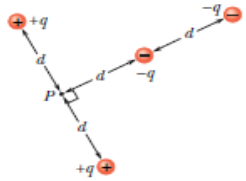
4. [72] The magnitude E of an electric field depends on the radial distance r according to $E= A/r^4$, where *A* is a constant with the unit volt–cubic meter. As a multiple of *A*, what is the magnitude of the electric potential difference between *r=2.00 m* and *r=3.00 m*?
5. [35] The electric potential at points in an *xy* plane is given by $V=(2.0\ V/m^2)x^2-(3.0\ V/m^2)y^2$. In unit-vector notation, what is the electric field at the point *(3.0 m,2.0 m)*?

6. [36] The electric potential V in the space between two flat parallel plates 1 and 2 is given (in volts) by $V = 1500x^2$, where x (in meters)is the perpendicular distance from plate 1. At x = 1.3 cm, (a) what is the magnitude of the electric field and (b) is the field directed toward or away from plate 1?

7. [14] Consider a point charge $q=1.0\ \mu C$, point A at distance $d_1=2.0m$ from q and point B at distance $d_2=1.0m$. *(a)* If A and B are diametrically opposite each other, as in *Figure-a,* what is the electric potential difference V_A-V_B? *(b)* What is that electric potential difference if A and B are located as in *Figure-b*?



8. [17] In the figure, what is the net electric potential at point P due to the four particles if V= 0 at infinity, q= 5.00 fC, and d= 4.00 cm?



9. [21] The ammonia molecule NH_3 has a permanent electric dipole moment equal to *1.47 D*, where *1 D = 1 debye unit = 3.34 × 10^{-30} Cm*. Calculate the electric potential due to an ammonia molecule at a point *52.0 nm* away along the axis of the dipole.(Set $V = 0$ at infinity.)
10. [13] What are (a) the charge and (b) the charge density on the surface of a conducting sphere of radius 0.15 m whose potential is 200 V(with V = 0 at infinity)?